Nematollah (Emad) Saeidi

AI/ML Engineer and Researcher

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Summary

AI researcher working on content-based medical image retrieval, graph neural networks, foundation models and histopathology image analysis.

Education

PhD **University of Isfahan**, Artificial Intelligence — Isfahan, Iran
Thesis: Content-based Medical Image Retrieval Using Graph Neural Networks. — Jan 2019 – Oct 2026

MSc **University of Isfahan**, Artificial Intelligence — Isfahan, Iran
Thesis: Learning Multi-Objective Binary Features for Image Representation and Recognition. — Jan 2015 – Jan 2017

BSc **Islamic Azad University Najaf Abad**, Software Engineering — Isfahan, Iran
Thesis: Intelligent Agent-based Social Simulation Model using Reinforcement Learning. — Jan 2010 – Jan 2013

Experience

Hyrcan Technology Development Co., AI Engineer — Tehran, Iran (Hybrid)
Nov 2025 – present
1 year

Developed a local RAG-based voice chat using LangChain, Ollama LLM, JSON/ChromaDB and Sentence Transformers, with few-shot prompting and conversation memory for dialogues in voice mode. Integrated Whisper (speech-to-text), a RAG-enhanced LLM and Coqui TTS into a full speech pipeline. Implemented real-time speaker recognition with NVIDIA NeMo TitaNet and Resemblyzer, including voice activity detection and acoustic feature extraction. Implemented real-time face recognition pipelines with YOLO for detection and ArcFace ResNet100 / InsightFace models for recognition.

Danube Private University Austria (DPU), MIAAI Group, AI Researcher — TFZ, Austria (Hybrid)
Sept 2023 – present
3 years 2 months

Histopathological image classification by nuclei-based graph convolutional networks and foundation models. Content-based medical image retrieval using graph neural networks. Image classification, graph neural networks, foundation models, histopathology image analysis, nuclei segmentation and generative models.

FANAP Co., Marketing Data Scientist — Tehran, Iran (Remote)
July 2021 – Sept 2023
2 years 3 months

Collaborated with cross-functional teams to troubleshoot and optimize the performance of 10+ deployed AI services. Designed and executed performance evaluations to assess model accuracy, speed and reliability, and identified and reported anomalies in AI systems. Drafted RFPs and developed business plans to support initiatives within the AI open innovation ecosystem. Authored technical articles and research briefs for the FANAP Research Team. Medical image search engine (Dec 2022 - Sep 2023): medical image retrieval and classification with an attention-based adversarial variational graph autoencoder, pre-trained models and approximate nearest neighbor search. Contour drawing for a robot motion planning system (Mar 2022 - Jul 2022): contour extraction with image processing techniques for a drawing robot.

FANAP Co., Data Scientist — Tehran, Iran (Hybrid)
Oct 2018 – July 2021
2 years 10 months

Worked in an agile environment with daily standups, sprint retrospectives, backlog grooming and continuous integration in two-week sprints. Programmed visual search and object detection AI services from scratch, and collaborated with the FANAP Research team on fraud detection, chatbot, video surveillance, recommendation and customer behavior analysis services. Object detection app with REST API (Sep 2020 - Jul 2021): detection and segmentation frameworks, few-shot models and transfer learning. Video surveillance-based attendance and registration app with REST API (Jan 2021 - Mar 2021). Automatic number plate recognition system (Jan 2021): vehicle registration, vehicle data search and display of top records by license plate. Visual search app with REST API (Jan 2020 - Sep 2020): object detection and recognition, color extraction, image web crawler, transfer learning and approximate nearest neighbor search. Value creation and clustering library (Aug 2019 - Jan 2020): SaaS metrics and KPIs, customer analysis and segmentation, cluster ensembles for the grocery retail industry. Customer behavior pattern mining on credit card transactions (Aug 2019 - Jan 2020) for fraud detection (money laundering, tax evasion). Recommendation system (Mar 2019 - Aug 2019): sequence-to-sequence ranking model, association rules, hybrid, content and collaborative filtering, customer clustering and feature engineering for candidate generation. Marketing analysis and product prediction (Oct 2018 - Feb 2019) with CatBoost, XGBoost, LightGBM, ensembles and MLP.

University of Isfahan, Research Assistant — Isfahan, Iran (On-site)
Oct 2019 – July 2023
3 years 10 months

Content-based medical image retrieval using graph neural networks. Image retrieval, graph neural networks, medical image analysis, generative models and foundation models.

University of Isfahan, Graduate Teaching Assistant — Isfahan, Iran (On-site)
Jan 2016 – June 2017
1 year 6 months

Machine Learning, Statistical Pattern Recognition and Algorithm Design courses under the supervision of Dr. Peyman Adibi.

Awards

Ernst Mach Grant, OeAD (Austrian Agency for International Cooperation in Education and Research) — Jan 2024
Research stay at Medical Image Analysis & Artificial Intelligence (MIAAI), Danube Private University (DPU), Austria.
OeAD

Ernst Mach Grant, OeAD (Austrian Agency for International Cooperation in Education and Research) — Jan 2023
Research stay at Medical Image Analysis & Artificial Intelligence (MIAAI), Danube Private University (DPU), Austria.
OeAD

Third place in the call to solve business challenges — Jan 2021
FANAP Co.

Second in the graduating class, MSc in Computer Engineering (Artificial Intelligence) — Jan 2017
University of Isfahan

Top two percent of graduated students, ranked 3rd among 158 BSc graduates in Computer Engineering — Jan 2013
Islamic Azad University Najaf Abad

Top Student, scholarships for two semesters — Jan 2011
Islamic Azad University Najaf Abad

Publications

Unfolding Graph Convolutional Networks: Architectures, Use Cases, and Trends Jan 2026
Nematollah Saeidi, Vahid Hajipour, S. Hekmat, B. Shoushtarian
doi.org/10.1016/j.asoc.2026.115122 (Applied Soft Computing (IF~8.2, Q1))

Evaluating Pre-trained Convolutional Neural Networks and Foundation Models as Feature Extractors for Content-based Medical Image Retrieval Jan 2025
Amirreza Mahbod, Nematollah Saeidi, S. Hatamikia, R. Woitek
doi.org/10.1016/j.engappai.2025.110571 (Engineering Applications of Artificial Intelligence (IF~8, Q1))

Leveraging Medical Foundation Model Features in Graph Neural Network-Based Retrieval of Breast Histopathology Images Jan 2025
Nematollah Saeidi, H. Karshenas, B. Shoushtarian, S. Hatamikia, R. Woitek, Amirreza Mahbod
(International Journal of Imaging Systems and Technology (IF~3.3, Q2))

Integrating Foundation Model Features into Graph Neural Network and Fusing Predictions with Standard Fine-Tuned Models for Histology Image Classification Jan 2025
Nematollah Saeidi, N. Torbati, R. Woitek, Amirreza Mahbod
doi.org/10.3390/bioengineering12121332 (Bioengineering, 12, 1332 (IF~3.7, Q2))

Incorporating Shape Dependent Power Law in Motion Planning for Drawing Robots Jan 2024
M. Abedinzadeh Shahri, Nematollah Saeidi, Vahid Hajipour
(Robotics and Autonomous Systems, 182, 104801 (IF~4.3, Q1))

Artificial Intelligence and Clinical Decision Making: Approaches and Challenges Jan 2022
Nematollah Saeidi, M. Torabi
(Journal of Applied Intelligent Systems and Information Sciences, 3(2), 43-55)

Attention-Based and Positional-Aware Neural Networks for Next-Item Recommendation Jan 2021
Nematollah Saeidi, H. Shahamat
(Journal of Applied Intelligent Systems and Information Sciences, 2(1), 29-40)

Single Sample Face Recognition Using Multicross Pattern and Learning Discriminative Binary Features Jan 2019
Nematollah Saeidi, H. Karshenas, H. M. Mohammadi
(Journal of Applied Security Research, 14(2), 169-190 (IF~1.2, Q2))

Learning Multi-Objective Binary Features for Image Representation Jan 2017
Nematollah Saeidi, H. Karshenas, H. M. Mohammadi
(2017 7th International Conference on Computer and Knowledge Engineering (ICCKE), Mashhad, pp. 47-52, IEEE)

Reviewer

Journals for which I serve as a peer reviewer

- Applied Soft Computing, Elsevier (Jul 2026 – present)
- Computer Methods and Programs in Biomedicine, Elsevier (Jun 2026 – present)
- Biomedical Signal Processing and Control, Elsevier (Sep 2024 – present)
- International Journal of Imaging Systems and Technology, Wiley (Sep 2026 – present)
- Cancer Control, SAGE
- Journal of Applied Intelligent Systems & Information Sciences (JAISIS), FANAP (May 2021 – present)

Skills

Generative AI and LLMs

Medical Image Analysis

Computer Vision

Machine Learning and Deep Learning

Programming and Tools

Data

Research and Professional

LinkedIn Skill Assessments passed

Languages

Persian
Native speaker

English
C1

German
A2

Interests

Medical Image Analysis

Graph Neural Networks

Applied AI

Certificates

Pretraining LLMs	Apr 2025
Finetuning Large Language Models	Mar 2025
Applied Plotting, Charting and Data Representation in Python	Oct 2022
Python Project for AI and Application Development	Sept 2022
Applied Machine Learning in Python	May 2022
Fifth IPM Advanced School on Computing and Artificial Intelligence	Sept 2021
Fourth IPM Advanced School on Computing and Artificial Intelligence	Aug 2020
IPM Advanced Data Science Summer School (ADS3)	Aug 2018
IPM Elementary Data Science Summer School (EDS3)	July 2018

Projects

Medical Image Retrieval with Foundation Models
Evaluation of pre-trained CNNs and medical foundation models as feature extractors for content-based medical image retrieval. Code: https://github.com/nematollahsaeidi/Medical-Image-Retrieval

Histology Image Classification with GNNs
Medical image classification using foundation model features in a graph neural network. Code: https://github.com/nematollahsaeidi/his_img_GNN_classification

Photo-Sketch
Contour generation from input images for drawing-robot motion planning. Code: https://github.com/nematollahsaeidi/Photo-Sketch

References

Amirreza Mahbod

Vahid Hajipour